

NUCULA v2 / JLCPCB

10 prototypes - 2026-09-21 - quotation / CAM review

Pending final supplier stackup, carrier and placement review

FABRICATION

4 layers; FR-4; 1.6 mm; black mask; white silk; ENIG. 1 oz outer / 0.5 oz inner. JLC04161H-3313. Electrical test required.

LAYER ORDER

F.Cu / In1.Cu / In2.Cu / B.Cu.

35 / 15.2 / 15.2 / 35 micrometre copper; 0.0994 mm surface prepreg; 1.265 mm core.

VIA FILL

Epoxy fill and copper cap all 0.20, 0.30 and 0.40 mm plated round holes (335). Includes U3 thermal holes. Leave 0.60 mm slots, 1.00 mm headers and all NPTH open.

ASSEMBLY

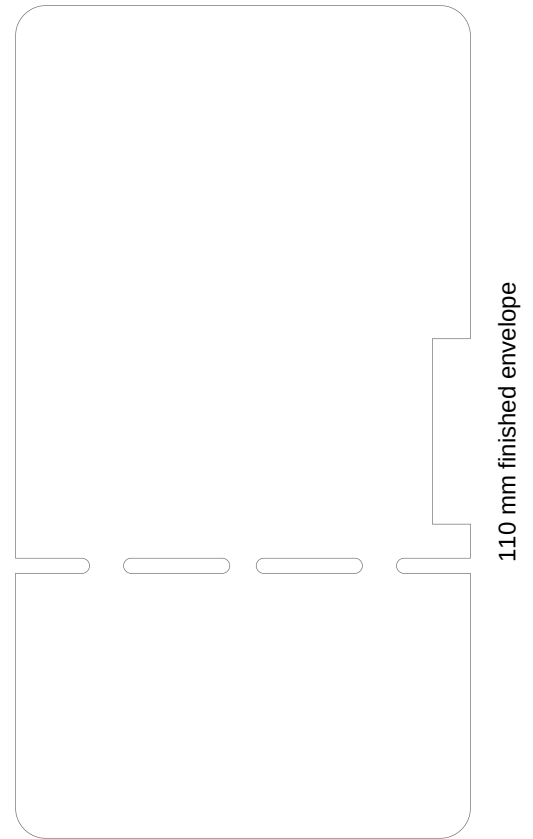
Top only. 116 placements, 55 BOM groups. Add supplier carrier, fiducials and tooling. Preserve ESP32 overhang. Keyboard remains attached.

USB CONNECTOR

J1 outer ground lands trimmed 0.04 mm at the hole-facing end. NPTH-to-pad clearance 0.233 mm; meets 0.20 mm minimum. Holes and shell stakes unchanged.

USB

90 ohm differential target, +/-10%; B.Cu referenced to In2.Cu; 0.15 mm traces / 0.21 mm gap. Confirm with actual stackup before production.



60 mm finished envelope

Outline from exported Gerber

FUNCTIONAL KEYBOARD - DO NOT FACTORY-SEPARATE

Lower keyboard: 60 x 35 mm. Routed gap: 2 mm. Two 4.5 mm support-tab necks.

28 NPTH mouse bites: 0.60 mm diameter; 1.00 mm pitch; 2 rows of 7 per tab.

Tab centers X=12 and 48 mm in export coordinates. Hole rows Y=34.7 and 37.3 mm.

Central electrical bridge: 3.5 mm wide; centered X=30 mm. Five traces cross it.

User cut at Y=36 mm is a drawing instruction only. Do not route or V-score here.

Drills: 354 plated round holes + 4 plated slots; 30 NPTH holes; 2 closed routed slots.

Origin: lower-left envelope corner. +X right, +Y up, millimetres. Top view.

DNP: C7 C8 C34 C35 C38 C39 J3 J4 J5 R38 R39. A1 / MB1 / MB2 are not parts.

Stencil: 0.10 mm nominal; U6 nine windows (61.91%); DS1 0.25 x 1.30 mm apertures.

See README.md for assembly preview checks, stock, and complete order remarks.